WHICH TOOL? MATCH WHAT THE QUESTION SAYS

Count arrangements, committees, seats, chips from a batch	Past paper: Counting and hypergeometric
Events per minute / per year, "no more than 2"	Past paper: Poisson, scale λ to the interval
"Which source is more likely to exceed..."	Past paper: $Q((t-\mu)/\sigma)$ for each, compare
Decide 0 or 1 from a noisy received value	Past paper: threshold from $f(y\|x)P(x)$ rule
A PDF has an unknown constant	Past paper: area = 1, then E, Var, CDF
Count successes in n independent trials	Binomial(n,p)
Trials until the first success	Geometric(p), memoryless
Waiting time, time to failure, arrivals	Exponential(λ), memoryless
Noise, measurement error, many small effects	Gaussian $N(\mu,\sigma^2)$, then standardise
Every value in a range equally likely	Uniform(a,b)
Probability at one exact value (continuous)	Always 0. Use an interval or the CDF
Add or average n measurements	Sum/mean rules; CLT for the shape
"As n grows, does the estimate converge?"	Law of Large Numbers
"Approximately normal for large n"	Central Limit Theorem
"Best guess" for a parameter	Point estimate ($\bar{x}$, s^2). Sample variance divides by n−1
"Plus or minus", "how confident"	Confidence interval $\bar{x} \pm z\cdot\sigma/\sqrt{n}$
"How many samples for ±E"	$n = (z\sigma/E)^2$, round up
Fit a line / plane through data	Linear regression, $w = X^{+}y$, add a column of 1s for bias
Curved data, choose degree	Polynomial features + BIC (take the minimum)
No closed form, iterative training	Gradient descent, $w \leftarrow w - \eta\nabla L$
Predict a class label (2 classes / K classes)	Sigmoid + BCE / softmax + cross-entropy
Overfitting, weights too large	Ridge: $w = (X^{T}X + \alpha I)^{-1}X^{T}y$

TOPICS

Exam format and marking (practice papers, mid-sem)

Use when: Read once before you start.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Practice papers say: open book, lecture notes and a calculator are permitted	Confirm for your own exam. The 2019 deferred paper allowed one A4 sheet.	Check the rules on your exam page. Do not assume unlimited notes.
The paper gives a Q-function table (Q(0.0) to Q(4.0)) and sometimes a Z-table	$Q(z) = 1 - \Phi(z)$.	Spend notes space on methods. You can look up numbers.
Section A: typed answers. Section B: handwritten. "No marks without reasoning, formulae or calculations"	Write the formula, substitute, then the number. A bare answer scores very little.	
Text box notation: a/b, a^(b-c), C(n,r), sqrt(a), Phi(x), $\text{sum}_{i=1}^{n}$		
Question mix: counting, Poisson or geometric, PDF with unknown constant, Gaussian tables, CI and sample size, joint PMF, regression and BIC, gradient descent, logistic and softmax, small NN	The AI practice exam has 30 short questions. Time per question is short.	

Exam

What the 2026 formula sheet and Z-table give you

Use when: Checked against the ECE2191 formula sheet and the Z-table.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Given: vector norm, inner product, cosine similarity		
Given: $W = (X^TX)^{-1}X^TY$ for multivariate regression, MSE, R^2, BIC, gradient descent update	Tall X (more samples than features).	The fat-matrix form $X^T(XX^T)^{-1}$ is NOT on the sheet.
Given: sigmoid, cross-entropy, softmax, sign, tanh, ReLU, MSE, BCE, NLL	Hinge loss and the log loss in practice 6 are not listed.	
Given: quadratic formula, product/quotient/chain rule, derivatives of x^n, e^x, ln, sin, cos, tan, hyperbolics	$\sigma' = \sigma(1-\sigma)$ is not listed. Derive it or keep it from this sheet.	
NOT given: any probability formula	No Binomial, Geometric, Poisson, Exponential, Gaussian PDF, CLT, CI, sample size, Bayes, covariance.	Put these first in your notes.
NOT given: ridge, precision, recall, F1, C(n+k,k) feature count, one-vs-rest, X^+ for fat matrices	All on this sheet.	
Z-table gives $\Phi(z) = P(Z \le z)$ to 4 decimals, for negative and positive z	$\Phi(-z) = 1 - \Phi(z)$. $Q(z) = 1 - \Phi(z)$.	Earlier practice papers gave a Q-table. The 2026 table is Φ, so subtract from 1 for tails.
Table values to expect: $\Phi(1)=0.8413$, $\Phi(1.5)=0.9332$, $\Phi(1.6)=0.9452$, $\Phi(1.96)=0.9750$, $\Phi(2)=0.9772$, $\Phi(2.8)=0.9974$	$Q(2)=0.0228$, $Q(1.5)=0.0668$, $Q(1.6)=0.0548$, $Q(2.8)=0.0026$ with the 4-decimal table.	
Row is z to one decimal, column is the second decimal	$\Phi(1.96)$: row 1.9, column 0.06.	Values can differ in the 4th decimal from software.

Exam Counting and hypergeometric

Use when: Seen: practice 4 Q1, practice 2 Q6, practice 3 Q3.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Round table, n people: $(n-1)!$ arrangements (fix one person)	Rotations identical.	Next to spouse among 20: $2 \cdot 18!/19! = 2/19$.
Choose k from n: $C(n,k) = n!/(k!(n-k)!)$	Order does not matter.	
Hypergeometric: P(exactly j faulty in k from N with M faulty) = $C(M,j)C(N-M,k-j)/C(N,k)$	Draw without replacement.	Chips: 30 total, 11 broken, test 5, exactly 1 faulty: $C(11,1)C(19,4)/C(30,5)$.
Sample space of the count of faulty chips among 5 drawn: $\{0,1,2,3,4,5\}$	Limited by how many faulty chips exist.	List values, not outcomes.
Groups by category: product of $C(\cdot,\cdot)$ for each category, divided by C(total, group size)	Clayton and Malaysia, male and female: 4 categories.	Check the chosen counts add to the group size.
Joint PMF from random draws (X reds, Y whites among 3 drawn from 2R, 3W, 4B): $P = C(2,x)C(3,y)C(4,3-x-y)/C(9,3)$	Marginals: sum the joint over the other variable.	X and Y are not independent (sample without replacement).

Stuck? Try: Write the total number of equally likely outcomes first, then count the favourable ones.

Exam Poisson counts

Use when: Seen: practice 2 Q1, practice 3 Q2. Not in your weekly notes, but on the papers.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$P(N=k) = e^{-\lambda}\lambda^k / k!$, $k=0,1,2,\ldots$ $E[N] = Var(N) = \lambda$	Events independent, constant average rate.	λ is for THE INTERVAL ASKED. Scale it first.
Rate scaling: λ = rate $\times$ interval length	Same time unit on both.	600/min over 200 ms: $\lambda = 600 \times (0.2/60) = 2$. 1460/year over one day: $\lambda = 4$.
$P(N=0) = e^{-\lambda}$	$\lambda=2 \Rightarrow 0.1353$. $\lambda=4 \Rightarrow 0.0183$.	
$P(N \le 2) = e^{-\lambda}(1 + \lambda + \lambda^2/2)$	Add the terms k=0,1,2.	$\lambda=2 \Rightarrow 5e^{-2} = 0.677$. $\lambda=4 \Rightarrow 13e^{-4} = 0.238$.
Time between Poisson events is Exponential(λ per unit time)	Link to the memoryless waiting-time model.	

Exam Gaussian tail comparisons and sample means

Use when: Seen: practice 2 Q2, practice 3 Q1, Q6, practice 4 Q7, mid-sem Q8.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$P(X > t) = Q((t-\mu)/\sigma)$	X Gaussian.	Source A (230, 5) vs B (225, 10) above 240: Q(2)=0.0228 vs Q(1.5)=0.0668. B more likely.
Lifetime ≥ 30,000: Sensor 1 (28000, 1000): Q(2)=0.0228. Sensor 2 (24000, 4000): Q(1.5)=0.0668	Choose the larger probability.	Choose Sensor 2.
Sample mean of n Gaussian readings: $\bar{X} \sim N(\mu, \sigma^2/n)$	Use $\sigma/\sqrt{n}$ as the SD.	Mid-sem Q8: $(13.2-12)/(6/\sqrt{64}) = 1.6 \Rightarrow 1-\Phi(1.6) = 0.0548$ (Z-table $\Phi(1.6)=0.9452$).
$P(45 < \bar{X} < 52)$, $\sigma=10$, $n=16$: SD = 2.5, z from 0 to 2.8	Q(0) − Q(2.8)	= 0.5 − 0.002555 = 0.4974.
Symmetric interval with probability p: $\mu \pm z\cdot\sigma/\sqrt{n}$ with $Q(z) = (1-p)/2$	95% ⇒ z=1.96. 60% ⇒ Q(z)=0.2 ⇒ z≈0.84.	95% of sample means: 45 ± 1.96(2.5) = (40.1, 49.9).
Required n so that $\lvert\bar{X} - \mu\rvert < \sigma/10$ with probability 0.95: $n = (1.96 \times 10)^2$	= 384.16 ⇒ n = 385. Round up.	
Peak height of a Gaussian PDF is $1/(\sigma\sqrt{2\pi})$: bigger peak = smaller σ	f(0) is a density, not P(A=0). P(A=0) = 0 for both.	

Exam PDF with an unknown constant

Use when: Seen: practice 2 Q3, practice 3 Q4, practice 4 Q2.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Total area = 1 fixes the constant	∫ f dx = 1 over the support.	f = a(4−(x−38)²) on [36,40]: area = a·32/3 ⇒ a = 3/32.
Second condition (e.g. E[X] given) fixes a second constant	f = a x^b on [0,1]: a = b+1, E = (b+1)/(b+2) = 0.6 ⇒ b = 0.5, a = 1.5.	
CDF: F(x) = ∫ from lower limit to x. Write 0 below and 1 above	Piecewise.	$F(x) = x^{1.5}$ on [0,1].
$P(X \geq c) = 1 - F(c)$	Symmetric PDF about 38 ⇒ P(X ≥ 38) = ½ and E = 38.	
$E[X] = \int x\,f$. $E[X^2] = \int x^2\,f$. $Var = E[X^2] - E[X]^2$	$f = 1.5x^{0.5}$: $E[X^2] = 3/7$, Var = 3/7 − 0.36 = 0.0686.	
$Y = aX + b$: $E[Y] = aE[X] + b$, $Var(Y) = a^2Var(X)$	Y = 2X+1: Var = 4Var(X). °F = 1.8·°C + 32: $Var_F = 3.24\,Var_C$.	
Piecewise PDF read from a graph: area = 1 gives a; Y = g(X) gives a PMF by summing probabilities of regions	P(X = 1) = 0 for continuous X.	

Exam Joint distributions: independent? uncorrelated?

Use when: Seen: practice 3 Q5, mid-sem Q3, Q15, practice 4 Q5.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Independence test: $P(x,y) = P(x)P(y)$ for every cell. One failing cell proves dependence	p(0,1)=p(1,0)=p(2,1)=1/3: P(X=0,Y=0)=0 but P(X=0)P(Y=0) = 1/9. Dependent.	
Uncorrelated test: $\text{Cov} = E[XY] - E[X]E[Y] = 0$	Same example: E[X]=1, E[Y]=2/3, E[XY]=2/3 ⇒ Cov = 0. Uncorrelated but dependent.	
X uniform on $\{-1,0,1\}$, $Y = X^2$: $E[XY] = 0$ ⇒ uncorrelated, but Y is a function of X	Zero covariance never proves independence.	
Conditional PDF: $f(y\|x) = f(x,y)/f_X(x)$, $f_X(x) = \int f(x,y)dy$	Fix x, integrate y over its limits at that x.	f = 2 on 0<y<x<1: $f_X = 2x$, $f(y\|x) = 1/x$, so $P(Y < x/2 \mid X=x) = (x/2)/x = ½$.
$E[aX+bY] = aE[X] + bE[Y]$. Cov(F, aF+b) = a Var(F)	Linearity of expectation needs no independence.	Practice 4 Q5: prove it from the integral definition.

Exam Bayes, total probability, event trees

Use when: Seen: practice 2 Q6c.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$P(A) = \Sigma\, P(A\|B_i)P(B_i)$ (total probability)	B_i partition the space.	P(not faulty) = 0.8·(5/30) + 0.6·(25/30) = 0.6333.
$P(B_i\|A) = P(A\|B_i)P(B_i)/P(A)$	P(new | not faulty) = 0.1333/0.6333 = 0.2105.	
Draw the tree and multiply along branches	Leaves sum to 1.	

Exam Binary detection with Gaussian noise

Use when: Seen: practice 4 Q3, practice 5 mixed.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Decide "0" if $f_Y(y\|X=-1)P(X=-1) > f_Y(y\|X=1)P(X=1)$	Y = X + N, N ~ N(0,σ²).	
Threshold: $T = (\sigma^2/2)\ln(P_0/P_1)$	Gaussian likelihoods with equal σ, means −1 and +1.	$P_0 = 2P_1$, $\sigma^2 = 1/16$ ⇒ $T = (1/32)\ln 2 = 0.0217$.
$P(\text{error}) = P_0\, Q((T+1)/\sigma) + P_1\, Q((1-T)/\sigma)$	σ = 0.25 here.	Q(4.09) and Q(3.91) are about 2×10^{-5} and 4.6×10^{-5}. Use the Q table.
Equal priors ⇒ $T = 0$	Unequal priors move the threshold toward the less likely symbol.	

Stuck? Try: Take logs of both sides, collect y on one side, and you get a straight threshold on y.

Exam

Concept traps from the mid-semester test

Use when: Multiple choice. Know why each wrong option is wrong.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Bursty errors: $S = \Sigma X_i$ with $P(X_i=1)=0.01$. $E[S] = 5$ still holds	Linearity needs no independence.	S is NOT Binomial and Var(S) is NOT np(1−p) without independence.
Geometric: 5 failures already; P(more than 3 more attempts) $= 0.8^3$	Memoryless.	Do not use 0.8^8 or 0.8^5.
CDF flat on (2.0, 2.5) ⇒ probability of that interval is 0	It does not mean uniform or discrete.	
CLT with n=30 and $P(X_i=1)=0.001$: $P(\bar{X}=0) = 0.999^{30} = 0.97$, so Gaussian approximation is poor	Rule n ≥ 30 is not a theorem. Very skewed parents need far larger n.	
LLN with a drifting offset D_i: not identically distributed, average drift need not vanish	IID is required.	More samples do not remove a systematic offset.
$\hat{X}_A = X_1$ and $\hat{X}_B = \bar{X}$ are both unbiased. $\hat{X}_B$ has SE $\sigma/\sqrt{n}$		
95% CI (−0.08, 0.12) inside tolerance ±0.20 supports "mean bias is within tolerance"	Not "95% of sensors" and not "95% of future errors".	
σ unknown, replace by S: interval follows Student t, not exact with z	Gaussian data.	
Halve the margin of error ⇒ quadruple n		
Exponential with mean 200 h: $\lambda = 0.005$, $\text{Var} = 1/\lambda^2 = 40{,}000$, $\text{SD} = 200$	Mean equals SD.	"Most fail near 200 h" is FALSE. The PDF is largest at 0 and decays. The bathtub (useful-life) remark does not answer the question.

Exam

Regression computations and properties (AI practice exam)

Use when: Seen: practice 6 Q1, Q3 to Q9, mock Q3 and Q6.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Residual $e_i = y_i - \hat{y}_i$. MSE $= (1/m)\Sigma e_i^2$	Least-squares fit with bias: $\Sigma e_i = 0$ and $\Sigma x_i\, e_i = 0$ and $\Sigma \hat{y}_i\, e_i = 0$.	
Recover lost residuals using $\Sigma e=0$ and $\Sigma xe=0$ (two equations, two unknowns)	Fitted by least squares with a bias.	Then $y_i = \hat{y}_i + e_i$.
Rescale the features: least squares gives the same predictions, ridge does not	Same α.	Ridge penalises $\|w\|^2$, which changes when units change.
$x'=2x-1$, $y'=10y+5$: $w' = 5w$, R^2 unchanged, MSE $\times$ 100	Affine changes of x and y.	$b' = 10b + 5w + 5$.
Fit y on x and x on y: $w_1 w_2 = R^2$, lines coincide only if $R^2=1$	S_xy, S_xx, S_yy. w_1 = S_xy/S_xx, w_2 = S_xy/S_yy.	Both lines pass through $(\bar{x}, \bar{y})$. SSE = S_yy − S_xy²/S_xx.
$R^2 = 1 - $ SSE/SST. Higher is better on training data	SST = $\Sigma(y-\bar{y})^2$.	R^2 on training data always improves with more features. Use validation or BIC.
Fewer samples than parameters: $w^\dagger = X^T(XX^T)^{-1}y$ (minimum norm)	Under-determined.	x=(1,2), y=5: $w^\dagger$ = (1,2). Prediction at (2,1) = 4, but another zero-error w=(5,0) predicts 10. Zero training error says little.
Features like cos x, sin x, x^2: still linear regression	Linear in the parameters.	Design matrix columns (1, cos x, sin x). With four equally spaced phases X^TX is diagonal.
Outliers pull the MSE line toward themselves	Spot large residuals, check the data, remove or use a robust loss, then refit.	
Polynomial features of degree $\le$ k from n inputs: C(n+k,k)	n=3: k=2 ⇒ 10, k=3 ⇒ 20.	20 parameters with 12 samples is under-determined. Training MSE ≈ 0. Do not trust it.

Exam BIC and model selection numbers

Use when: Seen: practice 6 Q2, Q10.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$\mathrm{BIC} = d \ln m + m \ln(\mathrm{MSE})$	Natural log. Lower is better.	m=50: A (d=3, 0.40) = −34.08. B (d=6, 0.33) = −31.96. A is selected.
Complexity term $d \ln m$ grows with degree. Fit term $m \ln \mathrm{MSE}$ falls	The sum has a minimum.	
B beats A only if $\mathrm{MSE}_B < 0.316$	Solve $6 \ln m + m \ln M < 3 \ln m + m \ln 0.40$.	
With $m = 5000$ the fit term dominates: B is selected	More data makes extra parameters easier to justify.	

Exam Gradient descent questions

Use when: Seen: practice 5 mixed, practice 6 Q12 to Q16.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Training loop: initialise w, repeat $w \leftarrow w - \eta \nabla L$ until a stopping rule	$\eta > 0$.	State the stopping rule: gradient small, loss change small, or max iterations.
$f = \frac{1}{2}x^2 - 4x + 12$: $\nabla f = x - 4$. One step to the minimum needs $\eta = 1$	Quadratic with curvature 1.	General: $f = (c/2)x^2$-type ⇒ best one-step $\eta = 1/\text{curvature}$.
$f(w) = (w-2)^2+1$: $w^{(t)} - 2 = (1-2\eta)^t(w^{(0)}-2)$	$w^{(0)}=5$, $w^{(1)}=3.8$ ⇒ $\eta=0.2$. Then 3.08, 2.648. $\eta=1$ oscillates 5, −1 forever.	
Convergence from every start: $\lvert 1 - \eta\lambda \rvert < 1$ for every curvature λ	$f = \frac{1}{2}(\omega_1^2 + 9\omega_2^2)$ ⇒ $0 < \eta < 2/9$.	The steepest direction limits the step size.
Learning rate choice: start moderate, watch the loss curve, decay when it plateaus	Too big: oscillate or diverge. Too small: slow.	A faster, still smooth curve after a change means η was raised.
Gradient of $f(x_1,x_2) = (x_1-3)^3\sigma(x_2)$: $\partial/\partial x_1 = 3(x_1-3)^2\sigma(x_2)$, $\partial/\partial x_2 = (x_1-3)^3\sigma(x_2)(1-\sigma(x_2))$	$\sigma' = \sigma(1-\sigma)$.	At (−1,1) ≈ (35.09, −12.58).

Exam Classification questions

Use when: Seen: practice 5 mixed, practice 6 Q17 to Q22.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Why not linear regression for classes: unbounded output, wrong loss, integer labels imply order	State at least two reasons.	
K classes with only logistic regression: K one-vs-rest models	3 classes ⇒ 3 models.	Pick the class with the largest probability.
Softmax uses exp: positive outputs, sum to 1. $z_i/\Sigma z$ fails with negatives or a zero sum		
Remove all activations from an MLP: it collapses to a single linear model	Composition of linear maps is linear.	
BCE per sample: $-y \ln \hat{y} - (1-y)\ln(1-\hat{y})$. Gradient $(\hat{y}-y)(x,1)$	$y \in \{0,1\}$.	The sample with the largest $\lvert\hat{y}-y\rvert$ has the largest gradient. Confident wrong predictions dominate.
Hinge loss $\max(0, 1-yf)$: for $y=-1$ minimised for any $f \le -1$ (greatest such value is -1). Log loss $\ln(1+e^{-yf})$: no finite minimiser		
Linear classifier $\text{sign}(w\cdot x+b)$: boundary $w_1x_1+w_2x_2+b=0$, w is normal to it	Side with $w\cdot x+b \ge 0$ is class +1.	$w=(2,-3)$, $b=1$: line $x_2 = (2/3)x_1 + 1/3$.
Metrics at a threshold: accuracy $(TP+TN)/N$, precision $TP/(TP+FP)$, recall $TP/(TP+FN)$	Raise the threshold: precision up, recall down.	If a missed fault is dangerous, choose the lower threshold (higher recall).
ReLU network: $f = \text{ReLU}(x) - 2\text{ReLU}(x-1) + \text{ReLU}(x-2)$ is a triangle with peak 1 at $x=1$, zero outside $[0,2]$	Piecewise linear, one kink per hidden unit.	

W1 Probability, random variables, expectation

Use when: Any "find the probability" or "find the mean" question.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$\Omega \to$ event $\to X \to$ PMF $\to E[X] \to \text{Var}(X)$	Write Ω first, then the event as a subset.	X=1 for "error" is not the transmitted bit.
$E[X] = \Sigma\, x\, P(X=x)$	PMF sums to 1.	Pair each probability with its own x.
$\text{Var}(X) = E[X^2] - (E[X])^2$, $SD = \sqrt{\text{Var}}$		
$E[aX+b] = aE[X]+b$, $\text{Var}(aX+b) = a^2\text{Var}(X)$	The shift b drops out of the variance.	

W1 Bernoulli, Binomial, Geometric

Use when: Counting successes or waiting for the first one.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Bernoulli(p): $P(1)=p$. $E=p$, $Var=p(1-p)$	One yes/no trial.	Var is largest at $p=½$.
Binomial(n,p): $P(X=k) = C(n,k)\, p^k\, (1-p)^{n-k}$. $E=np$, $Var=np(1-p)$	n fixed, independent trials, same p.	P(at least one) = $1-(1-p)^n$. E[X]=1 does not mean P(X≥1)=1.
Geometric(p): $P(X=k) = (1-p)^{k-1}\, p$, k=1,2,... $E=1/p$, $Var=(1-p)/p^2$	Counts trials including the first success.	Binomial: fixed n. Geometric: stop at first success.
Memoryless: $P(X>m+n \mid X>m) = P(X>n)$	Geometric only (discrete).	$P(X>n) = (1-p)^n$.

W1 Conditional, joint, independence

Use when: Event trees, tables of P(X,Y).

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$P(A\|B) = P(A,B)/P(B)$	P(B)>0.	$P(A\|B) \neq P(B\|A)$.
$P(A,B) = P(A\|B)P(B) = P(B\|A)P(A)$	Multiply along branches of an event tree.	
Total probability: $P(A) = P(A\|B)P(B) + P(A\|B^c)P(B^c)$	Denominator of Bayes sums over every branch.	
Bayes: $P(B\|A) = P(A\|B)P(B)/P(A)$		
Marginal: $P(X=x) = \Sigma_y\, P(X=x,Y=y)$		
Independent $\Leftrightarrow$ $P(x,y) = P(x)P(y)$ for ALL cells	Check every cell.	One mismatch settles it. One match proves nothing.

W1 Covariance, correlation, E[XY]

Use when: Two random variables together.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$E[XY] = \Sigma_x\, \Sigma_y\, xy\, P(x,y)$		
$E[X+Y] = E[X]+E[Y]$	Always. No independence needed.	
$Var(X+Y) = Var\, X + Var\, Y + 2Cov(X,Y)$	If independent, Cov=0.	Variance adds only for uncorrelated variables.
$Cov(X,Y) = E[XY] - E[X]E[Y]$		
$\rho = Cov(X,Y)/\sqrt{(Var\, X\, Var\, Y)}$, $-1 \le \rho \le 1$		
Independent ⇒ Cov=0. Cov=0 does NOT imply independent	Orthogonal means E[XY]=0. Uncorrelated means Cov=0.	
Model assumptions: independence, constant p, identically distributed, no memory	Bursts, fading, drift, wear break them.	Simple models underestimate rare extreme events.

W2 PDF and CDF

Use when: Continuous random variable questions.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$P(a \le X \le b) = \int f(x)dx$ from a to b	f ≥ 0 and total area 1.	P(X = x) = 0 for any single x.
$F(x) = P(X \le x) = \int f$ from $-\infty$ to x. $f(x) = dF/dx$		
$P(a<X\le b) = F(b) - F(a)$		
F is non-decreasing, $F(-\infty)=0$, $F(\infty)=1$	A PDF can exceed 1. A probability cannot.	
$E[X] = \int x\, f(x)dx$. $E[g(X)] = \int g(x)\, f(x)dx$		
$Var(X) = E[X^2] - (E[X])^2$		

Stuck? Try: Convert the question to an interval, then use the CDF.

W2 Uniform, Exponential, Gaussian

Use when: Pick the model from the story.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Uniform(a,b): $f = 1/(b-a)$. $F = (x-a)/(b-a)$ on [a,b]. $E = (a+b)/2$, $Var = (b-a)^2/12$	a ≤ x ≤ b.	Height is forced by area 1.
Exponential(λ): $f = \lambda e^{-\lambda x}$, x≥0. $F = 1 - e^{-\lambda x}$. $E = 1/\lambda$, $Var = 1/\lambda^2$	λ is a rate (events per unit time).	$P(X>x) = e^{-\lambda x}$. Models only the constant-failure-rate phase.
Exponential is memoryless: $P(X>s+t \mid X>s) = P(X>t)$	Continuous analogue of geometric.	
Gaussian $N(\mu,\sigma^2)$: $f = (1/(\sigma\sqrt{2\pi}))\, e^{-(x-\mu)^2/2\sigma^2}$	Many small independent effects.	No closed-form CDF. Standardise.
Empirical rule: $\mu\pm\sigma \approx 68.27\%$, $\mu\pm2\sigma \approx 95.45\%$, $\mu\pm3\sigma \approx 99.73\%$	Gaussian only.	

W2 Joint, marginal and conditional PDFs

Use when: Two continuous variables.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$P((X,Y) \text{ in region } R) = \iint f(x,y)\,dx\,dy$ over R	$f \geq 0$ and total volume 1.	Draw the region first, then set the limits.
Marginal: $f_X(x) = \int f(x,y)dy$	Limits can depend on x.	
Conditional: $f(x\|y) = f(x,y)/f_Y(y)$	$f_Y(y)>0$.	
Independent $\Leftrightarrow$ $f(x,y) = f_X(x)f_Y(y)$ for all x,y	Factorises, including the support.	Zero covariance alone is not enough.
$E[g(X,Y)] = \iint g\,f\,dx\,dy$. $Cov = E[XY] - E[X]E[Y]$		

W3 Sums and sample mean

Use when: Total or average of n readings.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$S_n = \Sigma X_i$. $E[S_n] = \Sigma E[X_i]$	Always.	
$Var(S_n) = \Sigma Var(X_i) + \Sigma\Sigma_{i\neq j}\, Cov(X_i,X_j)$	Independent: just $\Sigma Var(X_i)$.	Variance does not add automatically.
$\bar{X}_n = S_n/n$. $E[\bar{X}_n] = \mu$	Equal means.	
$Var(\bar{X}_n) = \sigma^2/n$, $SD = \sigma/\sqrt{n}$	IID (independent and identically distributed).	Averaging shrinks the estimator spread, not a single measurement.
$Var(cX) = c^2Var(X)$		

W3 Law of Large Numbers and Central Limit Theorem

Use when: Convergence versus shape of the distribution.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
LLN: $\bar{X}_n \to \mu$ as n grows	IID, finite mean and variance.	Says where, not how it is spread.
CLT: $Z_n = (\bar{X}_n - \mu)/(\sigma/\sqrt{n}) \to N(0,1)$	IID, finite mean and variance.	Fails with burst errors or drift (independence or identical distribution broken).
$\bar{X}_n \approx N(\mu, \sigma^2/n)$. $S_n \approx N(n\mu, n\sigma^2)$	n large. Rule of thumb $n \geq 30$.	Skewed or multimodal parents need larger n.
LLN: point. CLT: shape.		

W3 Standardising Gaussians, Φ and Q

Use when: Any P(X ≤ x) for a Gaussian.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$Z = (X-\mu)/\sigma \sim N(0,1)$	Subtract mean, divide by SD (not variance).	
$P(X \le x) = \Phi((x-\mu)/\sigma)$	Check whether the Z-table is lower-tail or upper-tail.	
$P(a<X<b) = \Phi((b-\mu)/\sigma) - \Phi((a-\mu)/\sigma)$		
$Q(z) = P(Z>z) = 1 - \Phi(z)$. $\Phi(-z) = 1 - \Phi(z)$	Q is the comms convention for tail probability.	
$\Phi(1.5) = 0.9332$. $\Phi(1)=0.8413$. $\Phi(1.96)=0.975$		

Stuck? Try: Standardise first. Sketch the bell and shade the area to decide Φ or 1−Φ.

W4 Point estimates and standard error

Use when: Estimate parameters from data.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$\hat{\mu} = \bar{x} = (1/n)\Sigma x_i$		
$\hat{\sigma^2} = s^2 = (1/(n-1))\ \Sigma(x_i-\bar{x})^2$	Population mean unknown, replaced by $\bar{x}$.	Divide by n−1, not n.
Fit a PMF: $\hat{p}$ = successes / trials. Fit a PDF: use $\hat{\mu}$, $\hat{\sigma^2}$	Then compute probabilities from the fitted model.	
$SE(\bar{X}) = \sigma/\sqrt{n}$ (or $s/\sqrt{n}$ if σ unknown)	More data shrinks SE, not the spread of one reading.	
Unbiased: $E[\hat{\theta}]=\theta$. Consistent: $\hat{\theta} \to \theta$ as n grows	They are different properties.	

W4 Confidence intervals

Use when: "Give a 95% interval", "plus or minus".

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$\bar{x} \pm z_{\alpha/2}\, \sigma/\sqrt{n}$	IID, σ known, population Gaussian or n large (CLT).	If σ is unknown the course flags Student t (beyond scope).
90%: $z = 1.645$. 95%: $z = 1.96$. 99%: $z = 2.576$		
Higher confidence ⇒ wider. Larger n ⇒ narrower.	Precision and coverage trade off.	
Meaning: 95% of intervals built this way contain μ	Repeated sampling.	It is not "95% probability μ is in this interval".
Total (sum) question: $S_n \approx N(n\mu, n\sigma^2)$	Do not use the sample-mean form for a total.	
Significance level = 100% − confidence level		

W4 Margin of error and sample size

Use when: "How many measurements?"

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$MOE = z_{\alpha/2}\, \sigma/\sqrt{n}$		
$n = (z_{\alpha/2}\, \sigma \,/\, MOE_max)^2$	σ known.	ALWAYS round up. Example: 61.47 → 62.
$n \propto 1/MOE^2$	Halving MOE needs 4× the data.	
Typical confidence: 90% exploratory, 95% default, 99% safety-critical		

W5 Vector random variables

Use when: $X = [X_1,\ldots,X_n]^T$, covariance matrices.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Mean vector $\mu = E[X]$		
Correlation matrix $R_{XX} = E[XX^T]$	Second moments.	Off-diagonals need not vanish under independence.
Covariance matrix $C_{XX} = E[(X-\mu)(X-\mu)^T]$	Diagonal = variances, off-diagonal = covariances.	
$C_{XX} = R_{XX} - \mu\mu^T$	Convert instead of recomputing every entry.	
C_{XX} is symmetric and positive semidefinite		
Gaussian vector: $f(x) = (1/((2\pi)^{n/2}\lvert C\rvert^{1/2}))$ $\exp(-\frac{1}{2}(x-\mu)^T C^{-1}(x-\mu))$	C nonsingular.	Check $\lvert C\rvert \neq 0$ first.
Jointly Gaussian and Cov=0 ⇒ independent	Both conditions.	Zero covariance alone does not give independence.
Contours: variances set spread along axes; covariance sets tilt	One state variable's uncertainty is the matching diagonal entry of C.	

W7 Univariate linear regression

Use when: Fit $\hat{y} = wx + b$ to data.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Zero-order model: $\hat{y} = b = \bar{y}$	Baseline. Good only if x and y are unrelated.	
Loss (MSE): $L = (1/m)\Sigma(y_i - wx_i - b)^2$	Convex: unique global minimum.	
Set $\partial L/\partial w = 0$ and $\partial L/\partial b = 0$ (Fermat)	Stationary point alone is not enough in general; convexity here guarantees a minimum.	
$w^* = \hat{\sigma_{xy}} / \hat{\sigma_{xx}}$, $b^* = \bar{y} - \bar{x}w^*$	$\hat{\sigma_{xy}} = (1/m)\Sigma(x_i-\bar{x})(y_i-\bar{y})$, $\hat{\sigma_{xx}} = (1/m)\Sigma(x_i-\bar{x})^2$	Divide by m, not m−1.
$w = (\Sigma x_i y_i - m\,\bar{x}\bar{y})/(\Sigma x_i^2 - m\,\bar{x}^2)$, $b = (\bar{y}\Sigma x_i^2 - \bar{x}\Sigma x_i y_i)/(\Sigma x_i^2 - m\,\bar{x}^2)$	Use when you are given raw sums.	
Supervised learning: data (x_i,y_i). Regression predicts a number, classification a label		
sklearn: LinearRegression().fit(X.reshape(-1,1), y); coef_, intercept_	X must be a column matrix (m,1).	

W8 Multivariate linear regression

Use when: Several inputs, matrix form.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$\hat{y} = w^T x$. X is m×n (one sample per row), y is m×1		
$w^* = X^+ y$	X full rank.	
m ≥ n (tall): $X^+ = (X^T X)^{-1} X^T$. m < n (fat): $X^+ = X^T (X X^T)^{-1}$	Fat case: infinitely many exact fits; X^+ gives the minimum-norm one.	
Bias: append a column of 1s to X; w_0 is the bias	Without it the plane passes through the origin (MSE can be much worse).	
Multiple outputs: $\hat{y} = W^T x$, $W^* = X^+ Y$ (Y is m×p)	Compute X^+ once for all outputs.	
$R^2 = 1 - \Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2$	R^2=1 perfect, R^2=0 same as the zero-order baseline.	
numpy: w = np.linalg.pinv(np.c_[np.ones(m), X]) @ y		

W9 Polynomial regression and overfitting

Use when: Nonlinear data, choosing the degree.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$\hat{y} = w_0 + w_1 x + ... + w_K x\text{^}K$ (Maclaurin truncation)	Linear in w. Same pseudo-inverse solution.	K is a hyperparameter.
Features of degree ≤ k from n inputs: C(n+k, k)	Includes cross terms, e.g. $x_1 x_2$, $x_1 x_2^2$. The count explodes (n=100, k=5: 96.5 million).	
Rule of thumb: at least 10 samples per parameter		
Training error falls as degree rises. Test error falls then rises	Underfit: both high. Overfit: train low, test high.	
BIC = d·ln(m) + m·ln(MSE)	d = number of parameters, m = samples, MSE on training data. ln is natural log.	BIC can be negative. Pick the degree with the minimum BIC.
sklearn: PolynomialFeatures(degree) in a pipeline with LinearRegression		

W10 Gradient descent

Use when: No closed form, or "do two iterations".

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$w^{(k)} = w^{(k-1)} - \eta \nabla f(w^{(k-1)})$	$\eta>0$ learning rate, f differentiable.	Update every coordinate with the OLD point, simultaneously.
$\nabla f = [\partial f/\partial x_1, ..., \partial f/\partial x_n]^T$ points to steepest ascent	Descent uses the negative gradient. Gradient is perpendicular to level sets.	
η too small: slow. Moderately large: bounces and plateaus. Too large: diverges		
Stop when $\|\nabla f\| \le \varepsilon$, or f decrease $\le \varepsilon$, or $\|w^{(k)} - w^{(k-1)}\| \le \varepsilon$, or $k = \max$		
Convex loss: reaches the global minimum. Non-convex (neural nets): a good minimum		
Direction-by-direction search needs 2^n checks	Does not scale. That is why the gradient is used.	
Chain rule: $y = \theta_3 + b$, $\theta_3 = w_2\theta_2$, $\theta_2 = \varphi(\theta_1)$, $\theta_1 = w_1 x$	$dy/dw_1 = w_2\, \varphi'(\theta_1)\, x$	Backpropagation is repeated chain rule.

Stuck? Try: Quadratic 1-D: $\nabla f = f'$, update $\omega \leftarrow \omega - \eta f'(\omega)$. Example $f=\omega^2+5\omega+6$, $\omega_0=2$, $\eta=0.1 \rightarrow 1.1$.

W11 Logistic regression

Use when: Two-class prediction.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$\sigma(z) = 1/(1+e^{-z})$, $\sigma(0)=0.5$, $\sigma' = \sigma(1-\sigma)$	$0 < \sigma < 1$.	Saturates for $\|z\| > 6$ (tiny gradient).
$\hat{y} = \sigma(w^T x)$. Predict class 1 if $\hat{y} \ge 0.5$	Bias via $x[0]=1$.	The decision boundary is a line or hyperplane.
BCE per sample: $\ell = -[y \ln \hat{y} + (1-y) \ln(1-\hat{y})]$	Natural log.	Do not use MSE with a sigmoid (vanishing gradient, wrong likelihood).
$\nabla_w L = (1/m) \Sigma (\hat{y}_i - y_i)\, x_i$	No closed form. Train by gradient descent. BCE is convex.	
Why linear regression fails: unbounded output, wrong objective, outliers skew the line		

W11 Softmax classifier

Use when: More than two classes.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$p_r = e^{z_r} / \Sigma e^{z_i}$	Probabilities sum to 1. Order of logits preserved.	
$z = W^T x$, W is n×K (one column per class)	Labels one-hot, not 1,2,3.	Integer labels imply a false ordering.
Loss: $L = -\Sigma y_r \ln \hat{y}_r = -\ln \hat{y}_c$	c = true class.	
$\nabla_W L = (1/m) \Sigma x_i(\hat{y}_i - y_i)^T$		
Example: z=(5.2, 3.6, −2.7) → p ≈ (0.832, 0.168, 0.0003)		

W11 Confusion matrix and metrics

Use when: Evaluate a classifier.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Rows = true class, columns = predicted. Diagonal = correct		
Accuracy = (TP+TN)/N	Misleading with imbalanced classes: always guessing the majority scores high.	
Precision = TP/(TP+FP). Recall = TP/(TP+FN)	Miss = FN. False alarm = FP.	
$F_1 = 2 \cdot P \cdot R/(P+R)$	Harmonic mean.	
TP=6, FP=2, FN=3 → P=0.75, R≈0.67, F_1≈0.71		
Imbalance fixes: class weights, resampling. predict_proba lets you change the 0.5 threshold		

W12 Ridge (L2) regression

Use when: Overfitting, many parameters, regularisation.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
$L_ridge = (1/m)\Sigma(\hat{y}_i - y_i)^2 + \alpha\|w\|^2$	α ≥ 0 hyperparameter.	α=0 gives ordinary least squares. Larger α shrinks w toward 0.
$w^* = (X^TX + \alpha I)^{-1} X^Ty$	Still has a closed form.	Adds α to the diagonal of X^TX.
Occam: prefer the simpler model that fits equally well		
sklearn: Ridge(alpha=1.0)	Tune α.	

W12 Neurons and shallow neural networks

Use when: Forward pass by hand, counting parameters.

FORMULA / FACT	CONDITIONS NEEDED	TRAP
Neuron: output $= \varphi(w^T x)$. $\varphi = \sigma$ gives logistic regression		
Hidden layer: $h = \varphi(W_1^T x)$. Output: $y = W_2^T h$	W_1 is n×H, W_2 is H×p. Bias via a 1 appended.	Apply φ elementwise.
ReLU: $\max(0,a)$. Sigmoid: $1/(1+e^{-x})$. tanh: $(e^x - e^{-x})/(e^x + e^{-x})$	ReLU range $[0,\infty)$. Sigmoid $(0,1)$. tanh $(-1,1)$.	ReLU can die (stuck at 0).
Classification head: $\text{softmax}(W_2^T h)$, cross-entropy. Regression: MSE	Pre-softmax values are logits.	
Parameters $\approx n \cdot H + H \cdot p$ (plus biases)	Example: 100·500 + 500·1 = 50,500.	
Universal approximation: one hidden layer, enough neurons, continuous f	More hidden units = more piecewise-linear joints.	
Training: backpropagation = chain rule + gradient descent		

Stuck? Try: Forward pass: multiply, apply ReLU (negative → 0), multiply again. Example gave y = 3.

Built from your ECE2191 weekly notes (W1 to W5, W7 to W12; no W6 supplied) and the practice papers 1 to 6 and mid-semester test. Poisson and counting come from the practice papers, not the notes. Closed-form variances for the uniform and exponential, and the Gaussian affine rules, are standard results added to complete the sheet; check them against your lecture notes.